UNIVERSITY OF THE PUNJAB

Roll No. .....

Course Code: CC-311-L **Time Allowed: 2 Hours**

Course: Operating Systems (Lab) **Max. Marks: 100**

Question # 1: **[25]**

A university laboratory maintains project files for different student groups on a Linux system.

Task: Write a **Linux shell script** to perform the following tasks:

1. Create a directory named StudentProjects.
2. Inside this directory, create three files: **group1.txt, group2.txt, and group3.txt**.
3. Display the details of all files using the command **ls -l**.
4. Modify the permissions of group1.txt so that:
   - The **owner** has read and write permissions.
   - The **group** has read permission.
   - **Others** have no permission.
5. Create both a **hard link** and a **symbolic link** for group2.txt.

Question # 2: **[25]**

A system administrator wants to monitor the creation of processes in Linux.

Write a **C program using system calls** to demonstrate process creation.
Display the process ID and parent process ID using **getpid()** and **getppid().** Create a child process using the **fork()** system call. Print different messages from the parent and child processes and use **wait()** so that the parent waits for the child process to finish execution.

Question # 3: **[25]**

Write a **multithreaded C program** using **POSIX threads** where three threads perform different tasks and display their thread IDs. Use **pthread_create()** to create the threads and **pthread_join()** to synchronize their execution.

Viva-Voce: **[25]**